Problem 1 (3pts + 3pts). *Prove or disprove the following equalities for arbitrary sets* A, B, C:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C). \tag{1}$$

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C). \tag{2}$$

Problem 2 (3pts + 3pts). 1. *Give the negation of the following statement* Q:

$$Q: \ \forall x \in \mathbb{Z}, \ (x \geq 100 \Rightarrow (\exists y \in \mathbb{Z} \text{ such that } y^2 = x)).$$

*Note: you don't need to prove if the statement is true or false. Just write down a statement that is the negation of Q.

2. *Are the following two formulas logically equivalent, justify your answer:*

$$(\neg(P \wedge Q)) \vee (P \wedge R) \quad \text{and} \quad (\neg P \vee \neg Q \vee R).$$

Problem 3 (6pts). *Prove that for any integer* $n \geq 1$,

$$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n+1)! - 1.$$

Problem 4 (4pts + 5pts + 3pts). *Suppose a coin is chosen uniformly at random from two coins: a fair coin A, and a biased coin B with* $\Pr(heads) = p > 1/2$. *The chosen coin is flipped n times independently.*

(a) *Let X denote the number of heads observed. Compute the posterior probability*

$$\Pr(Coin\ B \mid X = k).$$

(b) *Let P_n denote the posterior probability that the coin is biased after observing n consecutive heads. Note that $P_0 = 1/2$. Write a recurrence for P_{n+1} in terms of P_n.*

(c) *What is* $\lim_{n \to \infty} P_n$.

Problem 5 (Dynamic Programming). *A small inspection drone travels along a straight docking rail with n marked stations labeled $0, 1, \ldots, n-1$. The drone starts at station 0 and must reach the docking station $n-1$.*

At each station i, the drone's battery controller allows a single forward thruster burst whose maximum range is given by an integer **nums**$[i]$. *Concretely, if the drone is currently at station i, then in one burst it may move to any station*

$$i' \in \{i, i+1, \ldots, i + \textbf{nums}[i]\} \quad \text{such that} \quad i' < n$$

You may assume that station $n-1$ is reachable from station 0.

Formulate and implement a dynamic programming algorithm to Compute the minimum number of thruster bursts needed to reach station $n-1$ from station 0. Write the base cases, the recursive definition (with proper explanation) and analyze the running time.

Problem 6 (Graph Algorithms). *Consider a Snake and Ladder board consisting of N cells numbered $1, 2, \ldots, N$. The player starts at cell 1 and wants to reach cell N.*

Some cells are the base of a ladder, which immediately moves the player to a higher-numbered cell. Some cells are the mouth of a snake, which immediately moves the player to a lower-numbered cell. These moves happen automatically and do not consume an extra dice throw.

On each turn, the player rolls a die and may move forward by any number of cells between 1 and 6 (inclusive), as long as the destination cell number does not exceed N. The player has full control over the outcome of the die.

If the player lands on a cell with a snake or ladder, the corresponding move is applied immediately.

Design an algorithm to compute the minimum number of dice throws *required to reach cell N from cell 1.*

Problem 7 (Greedy Algorithms). *A server must broadcast a message to n clients. Client i requires $t_i > 0$ seconds of transmission time. The server can transmit to only one client at a time, and preemption is not allowed.*

Let C_i denote the time at which client i finishes receiving the message. The total latency *of a broadcast schedule is defined as*

$$\sum_{i=1}^{n} C_i.$$

Design an algorithm that schedules the transmissions so as to minimize the total latency. Prove that your algorithm is optimal.